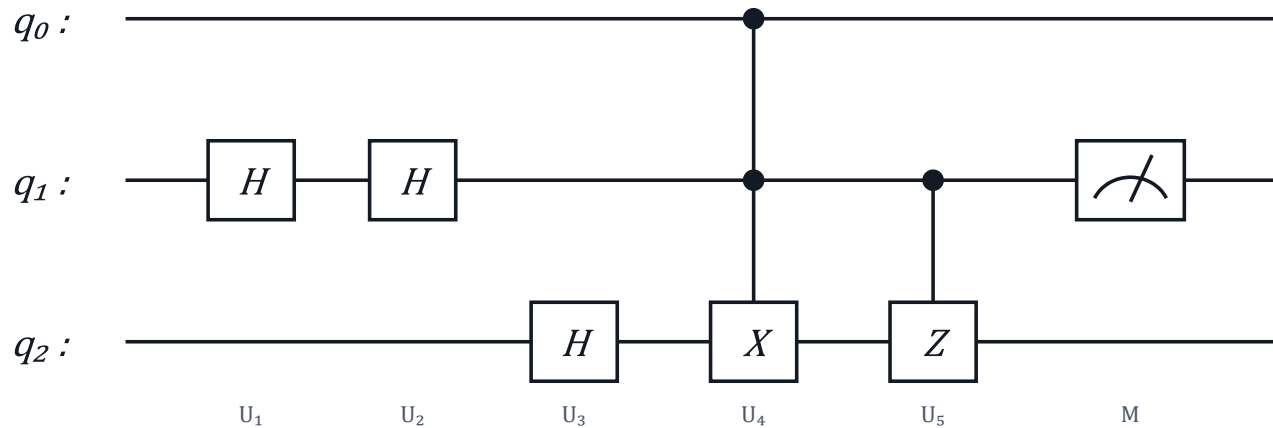


Álgebra del Circuito 2

Cómputo Cuántico · Simulador de circuitos cuánticos · Hajar González Fátima Estefanía



1. Convenciones

El estado del registro se escribe $|q_0q_1q_2\rangle$, con q_0 como bit más significativo. Un estado general es

$$|\psi\rangle = \sum_x \alpha_x |x\rangle, \quad x \in \{000, 001, \dots, 111\}, \quad \sum_x |\alpha_x|^2 = 1$$

La entrada del circuito es un estado producto $|\psi_0\rangle = |a\rangle \otimes |b\rangle \otimes |c\rangle = |abc\rangle$ con $a, b, c \in \{0, 1\}$. Por linealidad, el resultado para cualquier superposición de entrada se obtiene sumando los resultados de las entradas base con sus amplitudes.

Compuertas de un qubit

$$H = 1/\sqrt{2} \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} \quad X = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \quad Z = \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix}$$

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2} = |+\rangle \quad H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2} = |-\rangle$$

$$\text{En forma compacta: } H|b\rangle = (1/\sqrt{2})(|0\rangle + (-1)^b|1\rangle)$$

$$X|0\rangle = |1\rangle, X|1\rangle = |0\rangle \quad Z|0\rangle = |0\rangle, Z|1\rangle = -|1\rangle$$

$$X|+\rangle = |+\rangle, X|-\rangle = -|-\rangle \quad Z|+\rangle = |-\rangle, Z|-\rangle = |+\rangle$$

2. Operadores de cada paso

$$\begin{aligned}
 U_1 &= U_2 = I \otimes H \otimes I \quad (\text{Hadamard sobre } q_1) \\
 U_3 &= I \otimes I \otimes H \quad (\text{Hadamard sobre } q_2) \\
 U_4 &= CCX(q_0, q_1 \rightarrow q_2) = (I_8 - |11\rangle\langle 11| \otimes I) + |11\rangle\langle 11| \otimes X \\
 &\quad |a, b, c\rangle \mapsto |a, b, c \oplus ab\rangle \quad (\text{intercambia } |110\rangle \leftrightarrow |111\rangle) \\
 U_5 &= I \otimes CZ(q_1 \rightarrow q_2) = I \otimes (|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes Z) \\
 &\quad |a, b, c\rangle \mapsto (-1)^{bc} |a, b, c\rangle = \text{diag}(1, 1, 1, -1, 1, 1, 1, -1) \\
 |\psi_k\rangle &= U_k |\psi_{k-1}\rangle, \quad k = 1, \dots, 5
 \end{aligned}$$

Matriz de U_4 (Toffoli) en la base $|000\rangle \dots |111\rangle$

$$\begin{vmatrix}
 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
 \end{vmatrix}$$

3. Desarrollo general para la entrada $|abc\rangle$

Paso 1 · H sobre q_1

$$\begin{aligned}
 |\psi_1\rangle &= (I \otimes H \otimes I) |a\rangle |b\rangle |c\rangle = |a\rangle \otimes (1/\sqrt{2})(|0\rangle + (-1)^b |1\rangle) \otimes |c\rangle \\
 &= (1/\sqrt{2})(|a0c\rangle + (-1)^b |a1c\rangle)
 \end{aligned}$$

Paso 2 · H sobre q_1

$$\begin{aligned}
 |\psi_2\rangle &= |a\rangle \otimes (1/\sqrt{2})(H|0\rangle + (-1)^b H|1\rangle) \otimes |c\rangle \\
 &= |a\rangle \otimes (1/2)[(|0\rangle + |1\rangle) + (-1)^b (|0\rangle - |1\rangle)] \otimes |c\rangle \\
 &= |a\rangle \otimes (1/2)[(1 + (-1)^b)|0\rangle + (1 - (-1)^b)|1\rangle] \otimes |c\rangle \\
 &= |a\rangle |b\rangle |c\rangle = |abc\rangle \quad (\text{porque } H \cdot H = I)
 \end{aligned}$$

Paso 3 · H sobre q_2

$$|\psi_3\rangle = |a\rangle |b\rangle \otimes (1/\sqrt{2})(|0\rangle + (-1)^c |1\rangle) = (1/\sqrt{2})(|ab0\rangle + (-1)^c |ab1\rangle)$$

Paso 4 · Toffoli (controles q_0, q_1 ; objetivo q_2)

Si $ab = 0$ el estado no cambia. Si $a = b = 1$ se aplica X a q_2 :

$$\begin{aligned} X \cdot (1/\sqrt{2})(|0\rangle + (-1)^c|1\rangle) &= (1/\sqrt{2})(|1\rangle + (-1)^c|0\rangle) = (-1)^c \cdot (1/\sqrt{2})(|0\rangle + (-1)^c|1\rangle) \\ \Rightarrow |\psi_4\rangle &= (-1)^{abc} \cdot |a\rangle|b\rangle \otimes (1/\sqrt{2})(|0\rangle + (-1)^c|1\rangle) \end{aligned}$$

La Toffoli solo introduce una fase global -1 cuando $a = b = c = 1$ (porque $X|- \rangle = -|-\rangle$).

Paso 5 · CZ (control q_1 , objetivo q_2)

Si $b = 0$ el estado no cambia. Si $b = 1$ se aplica Z a q_2 :

$$\begin{aligned} Z \cdot (1/\sqrt{2})(|0\rangle + (-1)^c|1\rangle) &= (1/\sqrt{2})(|0\rangle - (-1)^c|1\rangle) = (1/\sqrt{2})(|0\rangle + (-1)^{c+1}|1\rangle) \\ \Rightarrow |\psi_5\rangle &= (-1)^{abc} \cdot |a\rangle|b\rangle \otimes (1/\sqrt{2})(|0\rangle + (-1)^{b \oplus c}|1\rangle) \\ &= (-1)^{abc} \cdot (1/\sqrt{2})(|ab0\rangle + (-1)^{b \oplus c}|ab1\rangle) \end{aligned}$$

Paso 6 · Medición de q_1

$$\begin{aligned} P(q_1 = k) &= \|(I \otimes |k\rangle\langle k| \otimes I)|\psi_5\rangle\|^2 \\ P(q_1 = b) &= |1/\sqrt{2}|^2 + |1/\sqrt{2}|^2 = 1, \quad P(q_1 = 1 - b) = 0 \\ \text{Estado tras medir: } |\psi'\rangle &= |\psi_5\rangle \quad (q_1 \text{ ya estaba en el estado definido } |b\rangle) \\ \text{Colapso del registro completo: } P(|ab0\rangle) &= 1/2, \quad P(|ab1\rangle) = 1/2 \end{aligned}$$

Resultado general. $|abc\rangle \rightarrow (-1)^{abc} \cdot |a\rangle \otimes |b\rangle \otimes H|b \oplus c\rangle$. Las dos Hadamard sobre q_1 se cancelan, por lo que la medición de q_1 devuelve siempre el valor inicial b con probabilidad 1. El qubit q_2 termina en $|+\rangle$ si $b = c$ y en $|-\rangle$ si $b \neq c$.

4. Ejemplos desarrollados

Entrada $|000\rangle$

$$\begin{aligned} |\psi_0\rangle &= |000\rangle = (1, 0, 0, 0, 0, 0, 0, 0)^T \\ |\psi_1\rangle &= (I \otimes H \otimes I)|000\rangle = (1/\sqrt{2})(|000\rangle + |010\rangle) = (1/\sqrt{2})(1, 0, 1, 0, 0, 0, 0, 0)^T \\ |\psi_2\rangle &= (1/\sqrt{2})[(1/\sqrt{2})(|000\rangle + |010\rangle) + (1/\sqrt{2})(|000\rangle - |010\rangle)] = |000\rangle \\ |\psi_3\rangle &= (I \otimes I \otimes H)|000\rangle = (1/\sqrt{2})(|000\rangle + |001\rangle) = (1/\sqrt{2})(1, 1, 0, 0, 0, 0, 0, 0)^T \\ |\psi_4\rangle &= CCX|\psi_3\rangle = (1/\sqrt{2})(|000\rangle + |001\rangle) \quad (q_0 = 0: \text{ la Toffoli no actúa}) \\ |\psi_5\rangle &= CZ|\psi_4\rangle = (1/\sqrt{2})(|000\rangle + |001\rangle) \quad (q_1 = 0: \text{ la CZ no actúa}) \\ \text{Medición de } q_1: P(q_1=0) &= 1/2 + 1/2 = 1, \quad P(q_1=1) = 0 \\ \text{Colapso: } P(|000\rangle) &= 50\%, \quad P(|001\rangle) = 50\% \end{aligned}$$

Entrada $|111\rangle$

$$|\psi_0\rangle = |111\rangle$$

$$|\psi_1\rangle = |1\rangle \otimes (1/\sqrt{2})(|0\rangle - |1\rangle) \otimes |1\rangle = (1/\sqrt{2})(|101\rangle - |111\rangle)$$

$$|\psi_2\rangle = (1/\sqrt{2})[(1/\sqrt{2})(|101\rangle + |111\rangle) - (1/\sqrt{2})(|101\rangle - |111\rangle)] = |111\rangle$$

$$|\psi_3\rangle = |11\rangle \otimes (1/\sqrt{2})(|0\rangle - |1\rangle) = (1/\sqrt{2})(|110\rangle - |111\rangle)$$

$$|\psi_4\rangle = \text{CCX}|\psi_3\rangle = (1/\sqrt{2})(|111\rangle - |110\rangle) = (1/\sqrt{2})(-|110\rangle + |111\rangle)$$

$$|\psi_5\rangle = \text{CZ}|\psi_4\rangle = (1/\sqrt{2})(-|110\rangle - |111\rangle) = -(1/\sqrt{2})(|110\rangle + |111\rangle) \quad (|111\rangle \text{ cambia de signo})$$

$$\text{Medición de } q_1: P(q_1=1) = 1/2 + 1/2 = 1, P(q_1=0) = 0$$

$$\text{Colapso: } P(|110\rangle) = 50\%, P(|111\rangle) = 50\%$$

5. Tabla completa para las 8 entradas base

$ \psi_0\rangle$	$ \psi_1\rangle$ (H q_1)	$ \psi_2\rangle$ (H q_1)	$ \psi_3\rangle$ (H q_2)	$ \psi_4\rangle$ (Toffoli)	$ \psi_5\rangle$ (CZ)	Medición q_1
$ 000\rangle$	$(1/\sqrt{2})(000\rangle + 010\rangle)$	$ 000\rangle$	$(1/\sqrt{2})(000\rangle + 001\rangle)$	$(1/\sqrt{2})(000\rangle + 001\rangle)$	$(1/\sqrt{2})(000\rangle + 001\rangle)$	$P(0) = 100\%$ $P(1) = 0\%$
$ 001\rangle$	$(1/\sqrt{2})(001\rangle + 011\rangle)$	$ 001\rangle$	$(1/\sqrt{2})(000\rangle - 001\rangle)$	$(1/\sqrt{2})(000\rangle - 001\rangle)$	$(1/\sqrt{2})(000\rangle - 001\rangle)$	$P(0) = 100\%$ $P(1) = 0\%$
$ 010\rangle$	$(1/\sqrt{2})(000\rangle - 010\rangle)$	$ 010\rangle$	$(1/\sqrt{2})(010\rangle + 011\rangle)$	$(1/\sqrt{2})(010\rangle + 011\rangle)$	$(1/\sqrt{2})(010\rangle - 011\rangle)$	$P(0) = 0\%$ $P(1) = 100\%$
$ 011\rangle$	$(1/\sqrt{2})(001\rangle - 011\rangle)$	$ 011\rangle$	$(1/\sqrt{2})(010\rangle - 011\rangle)$	$(1/\sqrt{2})(010\rangle - 011\rangle)$	$(1/\sqrt{2})(010\rangle + 011\rangle)$	$P(0) = 0\%$ $P(1) = 100\%$
$ 100\rangle$	$(1/\sqrt{2})(100\rangle + 110\rangle)$	$ 100\rangle$	$(1/\sqrt{2})(100\rangle + 101\rangle)$	$(1/\sqrt{2})(100\rangle + 101\rangle)$	$(1/\sqrt{2})(100\rangle + 101\rangle)$	$P(0) = 100\%$ $P(1) = 0\%$
$ 101\rangle$	$(1/\sqrt{2})(101\rangle + 111\rangle)$	$ 101\rangle$	$(1/\sqrt{2})(100\rangle - 101\rangle)$	$(1/\sqrt{2})(100\rangle - 101\rangle)$	$(1/\sqrt{2})(100\rangle - 101\rangle)$	$P(0) = 100\%$ $P(1) = 0\%$
$ 110\rangle$	$(1/\sqrt{2})(100\rangle - 110\rangle)$	$ 110\rangle$	$(1/\sqrt{2})(110\rangle + 111\rangle)$	$(1/\sqrt{2})(110\rangle + 111\rangle)$	$(1/\sqrt{2})(110\rangle - 111\rangle)$	$P(0) = 0\%$ $P(1) = 100\%$
$ 111\rangle$	$(1/\sqrt{2})(101\rangle - 111\rangle)$	$ 111\rangle$	$(1/\sqrt{2})(110\rangle - 111\rangle)$	$(1/\sqrt{2})(- 110\rangle + 111\rangle)$	$(1/\sqrt{2})(- 110\rangle - 111\rangle)$	$P(0) = 0\%$ $P(1) = 100\%$